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AI-DRIVEN

CYBER SECURITY

AI CAN WRITE CODE, ANALYZE THREATS, AND EVEN ACT LIKE A SECURITY ANALYST—BUT DO YOU ACTUALLY KNOW WHAT HAPPENS INSIDE AN LLM?

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WHAT IS AN LLM?

- LARGE LANGUAGE MODEL
- LEARNS PATTERNS FROM HUGE DATASETS
- PREDICTS THE NEXT TOKEN
- GENERATES HUMAN-LIKE TEXT
- USED IN AI APPLICATIONS

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TOKENS

- SMALL PIECES OF TEXT
- WORDS CAN BECOME MULTIPLE TOKENS
- MODELS PROCESS TOKENS, NOT SENTENCES
- INPUT AND OUTPUT USE TOKENS
- TOKENS AFFECT COST AND LIMITS

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CONTEXT WINDOW

- MODEL'S WORKING MEMORY
- HOLDS INPUT + CONVERSATION + INSTRUCTIONS
- MEASURED IN TOKENS
- LARGER CONTEXT = MORE INFORMATION
- LIMITED BY MODEL CAPACITY

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TRANSFORMERS

- CORE ARCHITECTURE BEHIND MODERN LLMS
- USES ATTENTION
- UNDERSTANDS RELATIONSHIPS BETWEEN TOKENS
- PROCESSES TEXT EFFICIENTLY
- ENABLES LARGE-SCALE LANGUAGE MODELS

ATTENTION

- FINDS IMPORTANT RELATIONSHIPS
- CONNECTS RELEVANT WORDS
- UNDERSTANDS CONTEXT
- KEY PART OF TRANSFORMERS

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HALLUCINATIONS

- AI GENERATES INCORRECT INFORMATION
- CAN SOUND VERY CONFIDENT
- NOT ALWAYS FACTUALLY GROUNDED
- DANGEROUS IN SECURITY
- REQUIRES VERIFICATION

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TEMPERATURE

- CONTROLS OUTPUT RANDOMNESS
- LOW TEMPERATURE → MORE CONSISTENT
- HIGH TEMPERATURE → MORE CREATIVE
- SECURITY TASKS → USUALLY LOWER TEMPERATURE
- DOES NOT GUARANTEE ACCURACY